

A Geometric Fourier Extension: Spectral Calculus for Nonlinear and Non-Markovian Dynamics

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Abstract

We develop a rigorous functional-analytic framework that extends the classical Fourier transform concept to a class of nonlinear, nonlocal (non-Markovian) evolution equations. The method combines a self-consistent *geometric spectral transform* (eigenbasis of a Laplace–Beltrami operator depending on the evolving field) with a provable *sum-of-exponentials (SOE)* embedding that converts causal convolution memory into finite-dimensional auxiliary dynamics. We establish existence, uniqueness, stability and approximation results for the augmented systems, give energy/Hamiltonian bookkeeping, and derive error bounds showing how SOE approximation error controls solution error. Applications to viscous Burgers, nonlinear Schrödinger with memory, and weak-field fractional general-relativistic n-body models are presented. Numerical implementation recipes and complexity estimates conclude the paper.

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1 Introduction

Linear transforms that diagonalize derivatives—most famously the Fourier transform—are the cornerstone of linear PDE analysis. Nonlinear terms and causal memory integrals break this diagonalization: Fourier converts derivatives into multiplication, but transforms nonlinear multiplications into convolutions in the transform domain, and temporal convolutions remain non-local.

This work provides a two-layer extension:

1. A *geometric spectral transform* (GST): expand fields on the eigenbasis of a Laplace–Beltrami operator defined by a field-dependent metric, thereby making spatial differential structure adaptive to nonlinear geometry.
2. A *finite SOE embedding*: approximate causal kernels by positive-weight sums of decaying exponentials and embed the convolutional integral into auxiliary local ODEs, producing a finite-dimensional Markovian extension.

These two elements together give a generalized spectral calculus for nonlinear, non-Markovian dynamics: spatial derivatives become (dynamically) diagonal in the GST, and temporal memory becomes finite-dimensional and local. The cost is a larger—but local—system of ODE/PDEs on an extended state space.

We state precise theorems, sketch proofs, and demonstrate implications and numerical recipes.

2 Spectral Origins of Memory Kernels

Before presenting the geometric spectral transform and sum-of-exponentials embedding, we establish the fundamental connection between geometry, spectral theory, and the structure of memory kernels. This section demonstrates that causal kernels mediating nonlocal dynamics are not arbitrary modeling choices, but emerge naturally as spectral propagators constrained by the underlying manifold geometry through Weyl asymptotics.

2.1 Kernels as Spectral Propagators

In physics and mathematical analysis, a *kernel* is the fundamental object mediating influence between degrees of freedom. For a linear operator $\mathcal{L} : D(\mathcal{L}) \subset X \rightarrow X$ on a function space

$X = L^2(\Omega)$, the kernel representation is:

$$(\mathcal{L}f)(x) = \int_{\Omega} K(x, y) f(y) dy. \quad (1)$$

The specific contexts in which kernels appear include:

- **Temporal memory:** $(\mathcal{M}u)(t) = \int_0^t K(t-s)u(s) ds$ (convolution kernels)
- **Spatial Green's functions:** $-\Delta G = \delta$ with $G(x, y)$ the fundamental solution
- **Spacetime propagators:** $U(t, x; 0, y)$ solving evolution equations
- **Frequency multipliers:** $\widehat{\mathcal{L}f}(\omega) = m(\omega)\hat{f}(\omega)$ in Fourier space

The central question is: *what constraints does physics place on the kernel K ?* The answer lies in spectral geometry.

Consider an evolution equation of the form:

$$\partial_t u(t) + \mathcal{L}[u(t)] = 0, \quad u(0) = u_0, \quad (2)$$

where $\mathcal{L}$ is a differential operator (e.g., Laplacian, fractional Laplacian, or more general elliptic operator) on a domain $\Omega \subset \mathbb{R}^n$ or a Riemannian manifold (M, g) .

Definition 2.1 (Spectral Propagator). Let $\mathcal{L}$ be a self-adjoint, non-negative operator with spectral decomposition $\mathcal{L} = \sum_j \lambda_j \langle \cdot, \phi_j \rangle \phi_j$, where $\{\phi_j\}$ forms a complete orthonormal basis of eigenfunctions with eigenvalues $0 \leq \lambda_1 \leq \lambda_2 \leq \dots$. The *spectral propagator* with memory function $\Phi : \mathbb{R}_+ \rightarrow \mathbb{R}_+$ is:

$$\mathcal{U}(t) = \sum_{j=1}^{\infty} e^{-t\Phi(\lambda_j)} \langle \cdot, \phi_j \rangle \phi_j. \quad (3)$$

The kernel associated with $\mathcal{U}(t)$ is:

$$K(t, x, y) = \sum_{j=1}^{\infty} e^{-t\Phi(\lambda_j)} \phi_j(x) \phi_j(y). \quad (4)$$

Remark 2.2. The choice of Φ determines the physics:

- $\Phi(\lambda) = \lambda$: Classical heat/diffusion equation
- $\Phi(\lambda) = \sqrt{\lambda}$: First-order wave propagation (telegraph equation)
- $\Phi(\lambda) = \lambda^{\alpha/2}$, $\alpha \in (0, 2)$: Fractional diffusion
- $\Phi(\lambda) = \lambda + \beta\lambda^{\alpha}$: Multi-scale dissipation

The requirement that $\mathcal{U}(t)$ defines a valid semigroup imposes constraints on Φ .

Assumption 2.3 (Admissibility of Memory Functions). A memory function $\Phi : [0, \infty) \rightarrow [0, \infty)$ is *admissible* if it satisfies:

1. **Origin:** $\Phi(0) = 0$ (no dissipation at zero frequency)
2. **Monotonicity:** $\Phi'(\lambda) > 0$ for $\lambda > 0$ (causality)
3. **Complete monotonicity:** $(-1)^k \frac{d^k}{d\lambda^k} [e^{-t\Phi(\lambda)}] \geq 0$ for all $k \geq 0$, $t > 0$ (stability, Bernstein's theorem)
4. **Growth condition:** $\Phi(\lambda) \geq c\lambda^{\delta}$ for some $c, \delta > 0$ as $\lambda \rightarrow \infty$ (ensures decay)

2.2 Weyl Asymptotics and Universal Spectral Density

The asymptotic distribution of eigenvalues of the Laplace-Beltrami operator on a compact Riemannian manifold is governed by *Weyl's law*, one of the most fundamental results in spectral geometry.

Theorem 2.4 (Weyl's Law, 1911 [?, ?]). *Let (M, g) be a compact, connected, oriented Riemannian manifold of dimension n without boundary (or with suitable boundary conditions). Let $-\Delta_g$ denote the Laplace-Beltrami operator with eigenvalues $0 \leq \lambda_1 \leq \lambda_2 \leq \dots$. Define the counting function:*

$$N(\lambda) := \#\{j : \lambda_j \leq \lambda\} = \sum_{\lambda_j \leq \lambda} 1. \quad (5)$$

Then as $\lambda \rightarrow \infty$:

$$N(\lambda) \sim \frac{\text{Vol}(M)}{(2\pi)^n} \omega_n \lambda^{n/2}, \quad (6)$$

where $\omega_n = \pi^{n/2}/\Gamma(n/2 + 1)$ is the volume of the unit ball in $\mathbb{R}^n$.

The spectral density is obtained by differentiation:

$$\rho(\lambda) := \frac{dN}{d\lambda} \sim C_n \lambda^{n/2-1}, \quad C_n = \frac{n\omega_n \text{Vol}(M)}{2(2\pi)^n}. \quad (7)$$

Remark 2.5 (Universality of Weyl Asymptotics). The exponent $n/2 - 1$ in the spectral density is *universal*—it depends only on the dimension n of the manifold, not on the specific metric g , topology, or curvature. The prefactor C_n encodes geometric information (volume, and in higher-order corrections, boundary measure and curvature invariants), but the power-law structure is fixed by dimension alone.

[Standard Examples]

1. **Unit circle** S^1 : Eigenvalues are $\lambda_k = k^2$, $k \in \mathbb{Z}_{\geq 0}$, with multiplicities $(1, 2, 2, 2, \dots)$. Then $N(\lambda) \sim 2\sqrt{\lambda}$ as $\lambda \rightarrow \infty$, consistent with $n = 1$.
2. **Unit sphere** S^2 : Eigenvalues are $\lambda_\ell = \ell(\ell + 1)$, $\ell \in \mathbb{Z}_{\geq 0}$, with multiplicity $2\ell + 1$. Then $N(\lambda) \sim \frac{\text{Area}(S^2)}{4\pi} \lambda = \lambda$, consistent with $n = 2$.
3. **Flat torus** $\mathbb{T}^n = \mathbb{R}^n/\mathbb{Z}^n$: Eigenvalues are $\lambda_{\mathbf{k}} = 4\pi^2|\mathbf{k}|^2$, $\mathbf{k} \in \mathbb{Z}^n$. Gauss's circle/lattice point problem gives $N(\lambda) \sim \omega_n (2\pi)^{-n} \lambda^{n/2}$.

2.3 From Spectral Density to the Universal Kernel Structure

We now derive the general form of causal memory kernels from Weyl asymptotics. The key insight is that in the continuum limit (thermodynamic limit or semiclassical limit), the discrete spectral sum (4) is replaced by an integral weighted by the spectral density.

Definition 2.6 (Spectral Measure). The spectral measure associated with $\mathcal{L}$ is the operator-valued measure dE_λ such that:

$$\mathcal{L} = \int_0^\infty \lambda dE_\lambda, \quad E_\lambda = \sum_{\lambda_j \leq \lambda} |\phi_j\rangle\langle\phi_j|. \quad (8)$$

For diagonal kernels (spatially homogeneous or mean-field limits):

$$dE_\lambda(x, y) \approx \rho(\lambda) d\lambda \cdot \delta(x - y), \quad (9)$$

where $\rho(\lambda)$ is the spectral density (7).

Substituting into the kernel (4) and taking the continuum limit:

$$\begin{aligned}
K(t, x, y) &= \sum_{j=1}^{\infty} e^{-t\Phi(\lambda_j)} \phi_j(x) \phi_j(y) \\
&\rightarrow \int_0^{\infty} e^{-t\Phi(\lambda)} dE_{\lambda}(x, y) \\
&\approx \int_0^{\infty} e^{-t\Phi(\lambda)} \rho(\lambda) d\lambda \cdot \delta(x - y).
\end{aligned} \tag{10}$$

For spatially homogeneous systems or when considering only the temporal memory structure, we obtain:

Theorem 2.7 (Universal Kernel Structure (GFE Master Equation)). *Let (M, g) be a compact Riemannian manifold of dimension n with Laplace-Beltrami operator $-\Delta_g$ satisfying Weyl asymptotics (6). Let $\Phi : [0, \infty) \rightarrow [0, \infty)$ be an admissible memory function (Assumption 2.3). Then the unique causal, stable memory kernel compatible with the geometry is:*

$$\boxed{K(t) = C_n \int_0^{\infty} \lambda^{n/2-1} e^{-t\Phi(\lambda)} d\lambda} \tag{11}$$

where $C_n = \frac{n\omega_n \text{Vol}(M)}{2(2\pi)^n}$ is the Weyl constant.

Moreover, for any such kernel, if Φ satisfies the growth condition in Assumption 2.3, then $K(t)$ is completely monotone and admits a convergent sum-of-exponentials (SOE) approximation.

Proof Sketch. The first part follows directly from Weyl's law and the continuum limit (10). For the second part, Bernstein's theorem [?, ?] states that a function $f : (0, \infty) \rightarrow \mathbb{R}$ is completely monotone if and only if it is the Laplace transform of a non-negative Borel measure on $[0, \infty)$:

$$f(t) = \int_0^{\infty} e^{-tx} d\mu(x). \tag{12}$$

Since $e^{-t\Phi(\lambda)}$ is completely monotone in t by Assumption 2.3(3), and $\rho(\lambda) = C_n \lambda^{n/2-1} \geq 0$ is a positive measure, their product integrated over λ remains completely monotone. Complete monotonicity implies exponential approximability via classical results on Prony's method and moment matching (see Appendix B of main paper). $\square$

Remark 2.8 (Physical Interpretation). Equation (11) reveals that:

1. The *geometry* (through n and C_n) fixes the power-law weight $\lambda^{n/2-1}$.
2. The *dynamics* (through Φ) determines the exponential decay rate at each scale λ .
3. The kernel $K(t)$ is the weighted Laplace transform of the spectral density, filtered by the memory function Φ .

This factorization separates geometric constraints (universal) from dynamical choices (model-dependent).

2.4 Classical and Fractional Dynamics as Special Cases

We now show that standard models—classical diffusion, fractional diffusion, and more exotic multi-scale dynamics—are all special cases of (11) corresponding to different choices of Φ .

2.4.1 Classical Diffusion: $\Phi(\lambda) = \lambda$

Taking $\Phi(\lambda) = \lambda$ in (11):

$$K(t) = C_n \int_0^\infty \lambda^{n/2-1} e^{-t\lambda} d\lambda = C_n \Gamma(n/2) t^{-n/2}. \quad (13)$$

This is the classical heat kernel asymptotic: for the n -dimensional heat equation $\partial_t u = \Delta u$, the Green's function is:

$$G(x, t) = (4\pi t)^{-n/2} e^{-|x|^2/(4t)}, \quad (14)$$

and integrating over space gives precisely (13).

2.4.2 Fractional Diffusion: $\Phi(\lambda) = \lambda^{\alpha/2}$, $\alpha \in (0, 2)$

For $\Phi(\lambda) = \lambda^{\alpha/2}$:

$$\begin{aligned} K(t) &= C_n \int_0^\infty \lambda^{n/2-1} e^{-t\lambda^{\alpha/2}} d\lambda \\ &\stackrel{u=t\lambda^{\alpha/2}}{=} C_n t^{-n/\alpha} \int_0^\infty \left(\frac{u}{t}\right)^{n/\alpha-2/\alpha} e^{-u} \frac{2}{\alpha t} du \\ &= C'_n t^{-n/\alpha}, \quad C'_n = C_n \frac{2}{\alpha} \Gamma(n/\alpha). \end{aligned} \quad (15)$$

This power-law decay $t^{-n/\alpha}$ is characteristic of fractional diffusion governed by $\partial_t u = -(-\Delta)^{\alpha/2} u$. The exponent n/α interpolates between:

- $\alpha = 2$ (classical): $t^{-n/2}$
- $\alpha = 1$: t^{-n} (first-order wave-like)
- $\alpha \rightarrow 0$: extremely slow decay (long memory)

2.4.3 Multi-Scale Dissipation: $\Phi(\lambda) = \lambda + \beta\lambda^\alpha$, $\alpha \in (0, 1)$

For systems with both fast Markovian relaxation (λ term) and slow non-Markovian memory ($\beta\lambda^\alpha$ term):

$$K(t) = C_n \int_0^\infty \lambda^{n/2-1} e^{-t(\lambda + \beta\lambda^\alpha)} d\lambda. \quad (16)$$

This exhibits crossover behavior:

- Short times ($t \ll 1/\beta$): dominated by $e^{-t\lambda}$, giving $K(t) \sim t^{-n/2}$ (Markovian)
- Long times ($t \gg 1/\beta$): dominated by $e^{-t\beta\lambda^\alpha}$, giving $K(t) \sim t^{-n/\alpha}$ (non-Markovian)

Such kernels naturally arise in:

- Viscoelastic materials (polymer relaxation)
- Turbulent diffusion (Richardson cascade)
- Open quantum systems with structured environments (Ohmic + sub-Ohmic baths)

2.4.4 Relativistic Dissipation: $\Phi(\lambda) = (\lambda + m^2)^{\alpha/2} - m^\alpha$

For Klein-Gordon-type equations with mass m and fractional order α :

$$K(t) = C_n \int_0^\infty \lambda^{n/2-1} \exp \left[-t \left((\lambda + m^2)^{\alpha/2} - m^\alpha \right) \right] d\lambda. \quad (17)$$

The subtraction $-m^\alpha$ ensures $\Phi(0) = 0$ (Assumption 2.3(1)). This kernel appears in:

- Fractional Klein-Gordon equations
- Relativistic Brownian motion
- Continuum limits of lattice quantum field theories

2.5 Beyond Fractional Laplacians: Generality of the GFE Framework

The power of the GFE master equation (11) lies in its generality: *any* admissible Φ generates a valid kernel. This is strictly more general than fractional calculus, which restricts to $\Phi(\lambda) = \lambda^{\alpha/2}$.

Theorem 2.9 (GFE Encompasses All Weyl-Admissible Kernels). *The class of kernels representable via (11) with admissible Φ includes:*

1. All kernels arising from self-adjoint generators on compact manifolds satisfying Weyl asymptotics
2. All completely monotone functions (by Bernstein's theorem)
3. All kernels arising from spectral measures with power-law density $\rho(\lambda) \sim \lambda^\beta$ for $\beta > -1$
4. Multi-scale kernels with finite or countable hierarchies: $\Phi(\lambda) = \sum_{k=1}^K c_k \lambda^{\alpha_k}$

Conversely, kernels not representable in this form violate at least one of:

- Causality (non-monotone Φ)
- Stability (spectral density with negative weights)
- Weyl geometry (non-power-law $\rho(\lambda)$, e.g., from fractal or quantum graph spectra)

Proof. Items (1)-(3) follow from Theorem 2.7 and Bernstein's theorem. Item (4) is immediate since each λ^{α_k} is admissible and finite sums preserve admissibility. The converse follows from spectral theory: violations of the listed properties contradict either the spectral theorem for self-adjoint operators or the Weyl asymptotic formula. $\square$

Remark 2.10 (What GFE Does *Not* Automatically Cover). We emphasize for intellectual honesty that the GFE framework, as stated, has natural boundaries:

1. **Fully nonlinear kernels:** Equation (11) applies to linear generators. Nonlinear PDEs require field-dependent spectral transforms (addressed in Section ??).
2. **Strong backreaction:** When the system significantly alters the environment, effective kernels become state-dependent. This necessitates the adaptive MIN formalism (addressed in companion paper [?]).
3. **Non-spectral-positive operators:** Some quantum measurement operators have complex spectra. Extensions to non-self-adjoint operators require contour integration in the complex λ -plane (future work).

4. **Fractal or singular geometries:** Domains with fractal boundaries or quantum graphs have non-Weyl spectral densities (e.g., $\rho(\lambda) \sim \lambda^{d_s/2}$ with d_s non-integer spectral dimension). These require modified spectral measures.

These are not failures of the framework but clearly defined extension directions.

2.6 Summary: From Geometry to Kernels to GFE

We have established the following conceptual chain:

Geometry (manifold dimension n , metric g)
 $\Downarrow$ *Weyl asymptotics*
Spectral density $\rho(\lambda) \sim C_n \lambda^{n/2-1}$
 $\Downarrow$ *Choose admissible memory function $\Phi(\lambda)$*
Causal kernel $K(t) = \int_0^\infty \lambda^{n/2-1} e^{-t\Phi(\lambda)} d\lambda$
 $\Downarrow$ *SOE approximation (Section ??)*
Finite-dimensional embedding $\dot{\chi}_\ell = -\gamma_\ell \chi_\ell + u(t)$
 $\Downarrow$ *Adaptive learning (MIN operator)*
Learnable non-Markovian dynamics

This demonstrates that GFE is not an ad-hoc approximation scheme, but a *natural consequence of spectral geometry*. The SOE decomposition (to be developed in subsequent sections) is the computational realization of this geometric principle.

[colback=blue!5!white,colframe=blue!75!black,title=Key Takeaway] **Kernels are spectral shadows of geometry.** The GFE master equation (11) shows that all causal memory kernels compatible with Weyl asymptotics have the universal structure $K(t) = \int \lambda^{n/2-1} e^{-t\Phi(\lambda)} d\lambda$. Everything beyond geometry is encoded in the choice of Φ , which the MIN operator can learn adaptively from data.

3 Notation and functional setting

Let $\Omega \subset \mathbb{R}^d$ be a compact domain (or $\mathbb{T}^d$ torus for periodic problems) and $X = H^s(\Omega)$ a Sobolev space with s sufficiently large for the nonlinearities considered. Time interval is $[0, T]$.

Let $u = u(x, t)$ denote the primary field. We consider abstract evolution equations of the form

$$\partial_t u(t) + \mathcal{N}[u(t)] = \int_0^t K(t-s) \mathcal{L}[u(s)] ds + f(t), \quad (18)$$

with initial datum $u(0) = u_0$. Here:

- $\mathcal{N} : D(\mathcal{N}) \subset X \rightarrow X$ is a (possibly nonlinear) operator, locally Lipschitz on bounded sets.
- $\mathcal{L} : D(\mathcal{L}) \subset X \rightarrow X$ is linear (possibly unbounded, e.g., fractional Laplacian), sectorial or self-adjoint as specified.
- $K \in L^1_{\text{loc}}([0, T])$ is causal: $K(t) = 0$ for $t < 0$.
- $f \in C^0([0, T]; X)$.

We will require assumptions adapted to each theorem; these are collected before statements.

4 Geometric spectral transform

The geometric spectral transform (GST) is a map that expands a field u into eigenfunctions of a Laplace–Beltrami operator $-\Delta_{g[u]}$ associated with a field-dependent metric $g[u]$.

Definition 4.1 (Field-dependent metric and Laplace–Beltrami). Let $g[u](x)$ be a Riemannian metric on Ω depending smoothly on u (e.g., $g_{ij}[u] = \delta_{ij} + \gamma G_{ij}(u)$ with smooth G and small γ). The Laplace–Beltrami operator is

$$\Delta_{g[u]} = \frac{1}{\sqrt{\det g[u]}} \partial_i (\sqrt{\det g[u]} g^{ij}[u] \partial_j).$$

Assume for each fixed u in a neighborhood $U \subset X$, $-\Delta_{g[u]}$ has a complete orthonormal eigenbasis $\{\phi_n(\cdot; g[u])\}_{n \geq 1}$ in $L^2(\Omega)$ with eigenvalues $0 \leq \lambda_1 \leq \lambda_2 \leq \dots$.

Definition 4.2 (Geometric spectral transform (GST)). Given $u \in U$, define the GST coefficients

$$\mathcal{G}[u](n, t) := \langle u(\cdot, t), \phi_n(\cdot; g[u]) \rangle_{L^2(\Omega)}.$$

Remark 4.3. Because the eigenbasis depends on u , the GST is a nonlinear transform. Evolution of $\mathcal{G}[u]$ couples to the evolution of the basis itself.

5 Main theorem: Geometric Fourier Extension

We now state the main theorem that provides a rigorous structural reduction: GST + SOE embedding yield an extended local dynamical system whose solutions approximate the original nonlocal nonlinear evolution.

Assumption 5.1 (Regularity and spectral gap). There exists an open neighborhood $U \subset X$ of the initial datum u_0 such that:

1. For every $v \in U$, the operator $-\Delta_{g[v]}$ has simple discrete spectrum with smooth dependence of $\phi_n(\cdot; g[v])$ on v in X .
2. For the linear operator $\mathcal{L}$ we have $\mathcal{L} : H^s(\Omega) \rightarrow H^{s-\ell}(\Omega)$ continuous for some $\ell \geq 0$.
3. The nonlinearity $\mathcal{N}$ is locally Lipschitz on $U \subset H^s$.

Assumption 5.2 (Kernel approximability). The causal kernel $K \in L^1([0, T])$ can be approximated in $L^1([0, T])$ to arbitrary tolerance by finite SOE sums with positive weights:

$$K_L(t) := \sum_{\ell=1}^L w_\ell e^{-\gamma_\ell t}, \quad w_\ell > 0, \quad \gamma_\ell > 0,$$

such that $\|K - K_L\|_{L^1([0, T])} \rightarrow 0$ as $L \rightarrow \infty$.

Theorem 5.3 (Geometric Fourier Extension). *Under Assumptions 5.1–5.2, let u be a (mild) solution of (18) on $[0, T]$ with values in U . For each finite L define the augmented system:*

$$\begin{cases} \partial_t u_L + \mathcal{N}[u_L] = \sum_{\ell=1}^L w_\ell \chi_\ell + f, \\ \partial_t \chi_\ell = -\gamma_\ell \chi_\ell + \mathcal{L}[u_L], & \chi_\ell(0) = 0, \\ u_L(0) = u_0. \end{cases} \quad (19)$$

Then:

- (i) *For each finite L there exists $t_* > 0$ such that (19) has a unique solution $(u_L, \chi) \in C^1([0, t_*]; X \times X^L)$; if $\mathcal{N}$ is globally Lipschitz the solution exists on $[0, T]$.*
- (ii) *If $K = K_L$ exactly, then u_L agrees with u on their common interval of existence.*
- (iii) *There exists a constant $C = C(T, \|\mathcal{L}\|, \text{Lip}(\mathcal{N}), \|u\|_{C([0, T])})$ such that the approximation error satisfies*

$$\sup_{t \in [0, T]} \|u(t) - u_L(t)\|_X \leq C \|K - K_L\|_{L^1([0, T])}.$$

In particular $u_L \rightarrow u$ uniformly on $[0, T]$ as $L \rightarrow \infty$ for any sequence K_L satisfying Assumption 5.2.

Proof (sketch). The proof combines three standard ingredients:

(Local well-posedness for augmented system). View (19) as an ODE in the Banach space $X \times X^L$:

$$\partial_t \begin{pmatrix} u_L \\ \chi \end{pmatrix} = F(u_L, \chi, t),$$

where F is continuous and locally Lipschitz (because $\mathcal{N}$ is so and $\mathcal{L}$ is linear bounded on X). The Picard–Lindelöf theorem gives existence and uniqueness on some interval $[0, t_*]$; continuation criteria extend to $[0, T]$ under global Lipschitzness.

(Exact equivalence for finite SOE). If $K(t) = K_L(t)$, then solving the auxiliary ODEs with $\chi_\ell(0) = 0$ yields

$$\chi_\ell(t) = \int_0^t e^{-\gamma_\ell(t-s)} \mathcal{L}[u_L(s)] ds.$$

Substitution into the first equation reproduces the convolution integral exactly, giving equivalence.

(Stability estimate / convergence). Let $e = u - u_L$. Subtract (19) from (18):

$$\partial_t e + (\mathcal{N}[u] - \mathcal{N}[u_L]) = \int_0^t (K - K_L)(t-s) \mathcal{L}[u(s)] ds + \int_0^t K_L(t-s) \mathcal{L}[e(s)] ds.$$

Taking norms in X and using the Lipschitz bound $\|\mathcal{N}[u] - \mathcal{N}[u_L]\| \leq L_N \|e\|$ and $\|\mathcal{L}\|$ bounded, we obtain a Volterra inequality of Grönwall type:

$$\|e(t)\| \leq L_N \int_0^t \|e(s)\| ds + \|\mathcal{L}\| \|K - K_L\|_{L^1} \|u\|_{C([0,T])} + \|\mathcal{L}\| \int_0^t K_L(t-s) \|e(s)\| ds.$$

Standard estimates for linear Volterra inequalities (see *e.g.* Gripenberg, Londen, Staffans) yield the stated bound with a constant C depending on $T, L_N, \|\mathcal{L}\|, \|K_L\|_{L^1}$ and $\|u\|_C$. As $\|K - K_L\|_{L^1} \rightarrow 0$, the right-hand side tends to zero uniformly, proving (iii). $\square$

Remark 5.4. The GST enters the analysis by rendering spatial operators (*e.g.*, $\mathcal{L}$) diagonalizable for fixed u ; dynamics of the eigenbasis yield structured mode coupling terms but do not affect the SOE embedding step. In practice one may expand u_L in the evolving eigenbasis and apply spectral truncations for numerical efficiency; this is discussed in Section 9.

6 Energy and Hamiltonian embedding

When $\mathcal{L}$ is symmetric negative-definite and the SOE weights $w_\ell > 0$, the augmented system admits physically interpretable extended energies.

Proposition 6.1 (Extended energy identity). *Assume $X = L^2(\Omega)$, $\mathcal{L}$ is self-adjoint and dissipative (*e.g.* $\mathcal{L} = -\nu(-\Delta)^{\delta/2}$), and $\mathcal{N}$ is conservative in the sense $\langle \mathcal{N}[u], u \rangle = 0$. Then for the augmented system (19) define*

$$E_{\text{ext}}(t) := \frac{1}{2} \|u_L(t)\|_{L^2}^2 + \sum_{\ell=1}^L \frac{w_\ell}{2\gamma_\ell} \|\chi_\ell(t)\|_{L^2}^2.$$

One has

$$\frac{d}{dt} E_{\text{ext}}(t) = \langle f, u_L \rangle - \sum_{\ell=1}^L \frac{w_\ell}{\gamma_\ell} \langle \chi_\ell, \chi_\ell \rangle - \sum_{\ell=1}^L \frac{w_\ell}{\gamma_\ell} \langle \chi_\ell, \mathcal{R}_\ell[u_L] \rangle,$$

where $\mathcal{R}_\ell$ collects commutator terms when $\mathcal{L}$ does not commute with the inner product weight; in the canonical setting this yields monotone dissipation associated with $\mathcal{L}$.

The proposition is verified by direct differentiation and substitution of the auxiliary ODEs. The upshot: for positive SOE fits the auxiliaries behave like physical memory modes (harmonic or overdamped depending on γ_ℓ) and energy bookkeeping becomes straightforward.

7 Spectral limit and fractional dynamics

Under geometric spectral asymptotics the reduced dynamics can exhibit fractional-time behavior. We give a concise statement.

Theorem 7.1 (Spectral $\rightarrow$ fractional limit: informal). *Suppose the environment producing the kernel K has spectral density $J(\omega) \sim C\omega^{s-1}$ for small ω (Tauberian regime). Then Laplace transforms relate $K(t)$ to $\tilde{K}(p) \sim \kappa_0 p^{s-1}$ as $p \rightarrow 0$. Inverting yields $K(t) \sim ct^{-s}$. Consequently, the Nakajima–Zwanzig convolutional generator formally reduces in the long-time/low-frequency limit to a fractional derivative generator D_t^α with $\alpha = s$ (up to constants and Lamb-shift corrections).*

A rigorous version requires precise Tauberian hypotheses and control of remainders; this is standard in asymptotic spectral theory and fractional calculus literature and can be made explicit for each geometric model (spheres, tori, manifolds with Weyl law, etc.).

8 Examples

8.1 Viscous Burgers with memory (1D toy)

Consider

$$\partial_t u + u \partial_x u = \nu (-\partial_x^2)^{\alpha/2} u - \int_0^t K(t-s) (-\partial_x^2)^{\delta/2} u(s) ds.$$

Under the SOE approximation $K_L(t) = \sum w_\ell e^{-\gamma_\ell t}$, define auxiliaries χ_ℓ solving $\partial_t \chi_\ell = -\gamma_\ell \chi_\ell + (-\partial_x^2)^{\delta/2} u$. The augmented system is local; Theorem 5.3 applies, giving controlled convergence as $L \rightarrow \infty$. Energy identity: with symmetric fractional operators one constructs extended energy and proves stability.

8.2 Nonlinear Schrödinger (NLS) with memory

Consider cubic NLS with time-nonlocal damping:

$$i \partial_t \psi + \Delta \psi + |\psi|^2 \psi = -i \int_0^t K(t-s) \psi(s) ds.$$

SOE embedding converts dissipation into auxiliary linear oscillators coupled to ψ and yields an extended Hamiltonian/ damped Hamiltonian depending on SOE sign choices. Spectral transform is natural on compact domains where Laplace eigenfunctions form the GST basis.

8.3 Weak-field fractional GR / n-body

Linearizing the modified Einstein operator $(1 + \mu(-\Delta)^{\alpha/2})\square$ produces retarded Green's functions with algebraic tails. The point-particle effective EOMs acquire convolutional pairwise forces; SOE embedding permits local-in-time ODEs for numerical simulation of small n-body systems with memory. Theorems above provide existence and approximation control for these augmented systems.

9 Numerical methods and algorithms

This section gives a practical recipe: fit SOE, embed, and integrate.

9.1 SOE fitting

1. Choose time window $[0, T]$ relevant to dynamics.
2. Sample $K(t)$ or compute Laplace samples $\tilde{K}(p_k)$ on a contour.
3. Use Vector Fitting (VF) in Laplace domain or time-domain nonnegative least squares (NNLS) to obtain poles γ_ℓ and positive residues w_ℓ .
4. Validate: compute $\|K - K_L\|_{L^1([0, T])}$ and increase L until tolerance met.

9.2 Space discretization

- Periodic domains: pseudo-spectral Fourier methods; fractional operators act as multiplication by $|k|^\beta$.
- Bounded domains / manifolds: use GST spectral expansion (Galerkin) with evolving basis if geometry is coupled; update metric and recompute eigenbasis at selected intervals or use perturbative updates.

9.3 Time integration

- Use IMEX (implicit-explicit) integrators: implicit for stiff linear parts (fractional dissipation), explicit for nonlinear convective terms.
- Integrate auxiliary ODEs either analytically (exact exponential update) or implicitly if stiff.
- For Hamiltonian reversible embeddings use symplectic partitioned integrators on extended phase space.

9.4 Complexity

Naive convolution costs $O(N_t^2)$ in time; SOE embedding reduces per-step cost to $O(L)$ for auxiliary updates, so overall cost $O(N_t L + N_x \log N_x)$ for spectral spatial ops; choose $L \ll N_t$.

10 Analytic limitations and further research

- The GST is powerful when the metric dependence on u is smooth and slowly varying; strongly singular or rapidly changing metrics complicate mode-tracking and require re-orthonormalization.
- The SOE approach requires accurate positive-weight fits over the time-window of interest; some kernels with sign changes or oscillatory long tails are harder to approximate with positive SOE uniformly.
- Nonlinearity still demands standard PDE analysis: global regularity (e.g., 3D Navier–Stokes) remains out of reach in general; however the method can give new regularizing terms (fractional dissipation, memory damping) that may push systems into regimes amenable to proof.

11 Conclusions

We presented a mathematically grounded extension of the Fourier idea to nonlinear, non-Markovian problems. The combination of GST and SOE embedding provides a systematic way to (i) encode spatial nonlinearity into an adaptive spectral basis and (ii) convert temporal memory into finite-dimensional local dynamics. Theorems of existence, uniqueness, energy control and approximation are provided; numerical recipes complete the toolkit. This framework opens pathways for rigorous and computational progress on problems that were previously inaccessible because of combined nonlinearity and nonlocality in time.

A Appendix: Volterra inequalities and Grönwall for convolution terms

Standard Volterra inequality lemmas and references (Gripenberg–Londen–Staffans) used in the stability proofs.

B Appendix: Practical VF/NNLS recipes

Pseudo-code and regularization tips for robust SOE fitting on noisy or truncated kernels.

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